

C vs. C++

Feature	C	C++
Programming Paradigm	Procedural (focus on functions and step-by-step instructions)	Object-Oriented + Procedural (supports classes, objects, inheritance, polymorphism, etc.)
Approach	Top-down approach	Bottom-up approach
Data Security	Less secure (no strong encapsulation)	More secure (supports encapsulation using classes)
Functions	Does not support function overloading	Supports function overloading
Data Types	Does not support reference variables	Supports reference variables
Memory Management	Uses malloc(), calloc(), free()	Uses new, delete (along with C functions)
Input/Output	Uses printf() and scanf()	Uses cin and cout (from iostream)
Standard Library	Smaller standard library	Larger standard library (includes STL: vectors, maps, lists, etc.)

Basic io

C

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter a
number: ");
    scanf("%d", &num);
    printf("You entered:
%d", num);
    return 0;
}
```

C++

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a
number: ";
    cin >> num;
    cout << "You entered:
" << num;
    return 0;
}
```

```
using namespace std;
```

```
#include <iostream>
using namespace std;
```

```
int main() {
    cout << "Hello
World";
    return 0;
}
```

```
#include <iostream>
```

```
int main() {
    std::cout << "Hello
World";
    return 0;
}
```

while loop

C

```
#include <stdio.h>
int main() {
    int i = 1;
    while(i <= 5) {
        printf("%d ", i);
        i++;
    }
    return 0;
}
```

C++

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    while(i <= 5) {
        cout << i << " ";
        i++;
    }
    return 0;
}
```

do while

C

```
#include <stdio.h>
int main() {
    int i = 1;
    do {
        printf("%d ", i);
        i++;
    } while(i <= 5);
    return 0;
}
```

C++

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    do {
        cout << i << " ";
        i++;
    } while(i <= 5);
    return 0;
}
```

for loop

C

```
#include <stdio.h>
int main() {
    int i;
    for(i = 1; i <= 5;
i++) {
        printf("%d ",
i);
    }
    return 0;
}
```

C++

```
#include <iostream>
using namespace std;
int main() {
    for(int i = 1; i <=
5; i++) {
        cout << i << "
";
    }
    return 0;
}
```

Print diamond pattern of stars (*)

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter number of
rows: ";
    cin >> n;
    // Upper part
    for(int i = 1; i <= n;
i++) {
        for(int s = 1; s <=
n-i; s++) cout << " ";
        for(int j = 1; j <=
2*i-1; j++) cout << "*";
        cout << endl;
    }
```

```
    // Lower part
    for(int i = n-1; i >= 1;
i--) {
        for(int s = 1; s <=
n-i; s++) cout << " ";
        for(int j = 1; j <=
2*i-1; j++) cout << "*";
        cout << endl;
    }
    return 0;
}
```


Program: Electricity Bill Calculator (with menu + multiple customers)

```
#include <iostream>
using namespace std;
int main() {
    int choice;
    char again;
    do {
        int units;
        float bill = 0;
        cout << "\n===== ELECTRICITY BILL\n";
        MENU =====\n";
        cout << "1. Domestic Consumer\n";
        cout << "2. Commercial Consumer\n";
        cout << "3. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
                cout << "Enter units consumed: ";
                cin >> units;
                if (units <= 100)
                    bill =
```

```
else if (units <= 300)
    bill = (100 * 5) + (units - 100) * 7;
    else
        bill = (100 * 5) + (200 * 7) + (units - 300) * 10;
        cout << "Your Electricity Bill (Domestic): Rs. " << bill << endl;
        break;
        case 2:
            cout << "Enter units consumed: ";
            cin >> units;
            if (units <= 200)
                bill = units * 8;
            else if (units <= 500)
                bill = (200 * 8) + (units - 200) * 12;
            else
                bill = (200 * 8) + (300 * 12) + (units - 500) * 15;
            cout << "Your Electricity Bill (Commercial): Rs. " << bill << endl;
            break;
            case 3:
                cout << "Goodbye!\n";
                return 0;
            default:
                cout << "Invalid Option!\n";
        }
        cout << "Calculate bill for another customer? (y/n): ";
```

Countdown timer with clear screen

```
#include <iostream>
#include <unistd.h>    //
for sleep()           //
#include <cstdlib>      //
for system()          //
using namespace std;
int main() {
    int seconds;
    cout << "Enter time
in seconds: ";
    cin >> seconds;

    while (seconds > 0) {
        system("clear");
        // clear screen
        cout << "Time
left: " << seconds << "
seconds\n";
        sleep(1);
        // wait 1 second
        seconds--;
    }
    system("cls");
    cout << "Time's up!"
<< endl;
    return 0;
```

Countdown timer with progress bar

```
#include <iostream>
#include <unistd.h>    // for
usleep()
using namespace std;
int main() {
    int seconds;
    cout << "Enter time in seconds:
";
    cin >> seconds;
    int totalSteps = seconds * 10;
    // 10 updates per second
    for(int step = 0; step <=
totalSteps; step++) {
        float percent = (step /
(float)totalSteps) * 100;
        cout << "\r[";

        // Print filled:
        for(int i = 0; i < step;
i++)
            cout << ":";
        // Print remaining:
        for(int i = step; i <
totalSteps; i++)
            cout << " ";
        cout << "]" << (int)percent
<< "% " << flush;

        usleep(100000); // 0.1
second delay
    }
    cout << "\nTime's up!" << endl;
    return 0;
}
```

Write a C++ program to create an ATM menu using switch and do-while. Allow user to check balance, deposit money, withdraw money (with sufficient balance check), and exit.

```
#include <iostream>
using namespace std;
int main() {
    int balance = 10000, choice, amount;
    char again;
    do {
        cout << "\n===== ATM MENU =====\n";
        cout << "1. Check Balance\n";
        cout << "2. Deposit Money\n";
        cout << "3. Withdraw Money\n";
        cout << "4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
                cout << "Current Balance: " <<
                    balance << endl;
                break;
            case 2:
                cout << "Enter amount to
                    deposit: ";
                cin >> amount;
                balance += amount;
                cout << "Amount Deposited.\n";
                break;
            case 3:
                cout << "Enter amount to withdraw: ";
                cin >> amount;
                if (amount <= balance)
                    balance -= amount;
                else
                    cout << "Insufficient Balance.\n";
                break;
            case 4:
                cout << "Goodbye!\n";
                return 0;
            default:
                cout << "Invalid Choice.\n";
        }
        cout << "Do you want to perform another
            transaction? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    return 0;
}
```